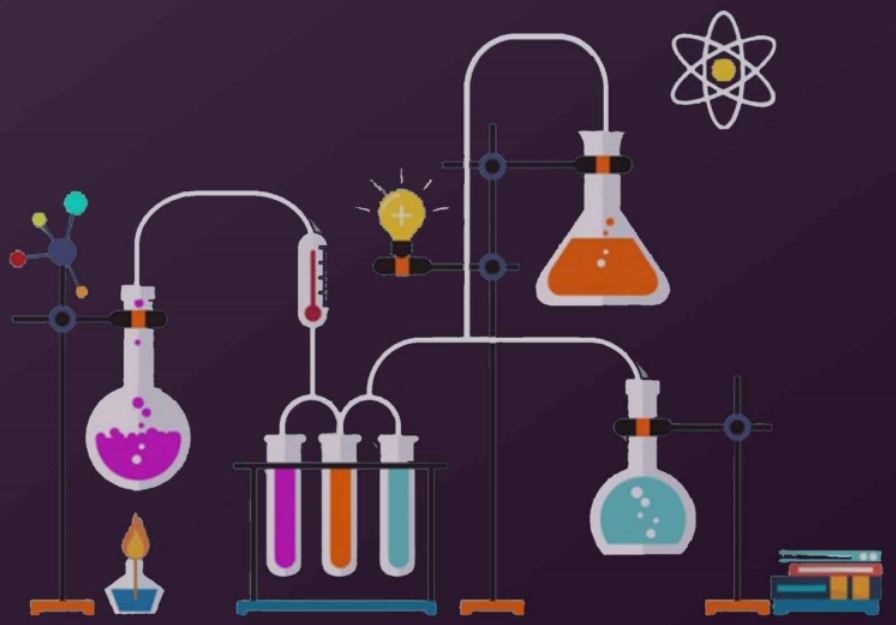
CHEMISTRY

# General Organic Chemistry



## SHEET

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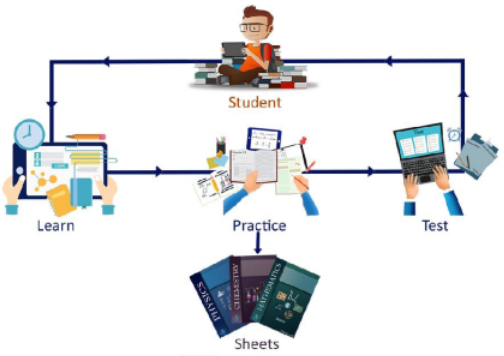
# *Contents*


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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr. No. | Topics | Ex-1 | Ex-1A | Ex-2 | Ex-2A |
|---|---|---|---|---|---|
| 1 | Introduction to Electronic Displacement Effects | | | | |
| 2 | Basics of General Organic Chemistry | | | | |
| 3 | Inductive Effect | 1-11 | 1 | 1 | |
| 4 | Acidity and Basicity | 12-20 | 2 | 2-16 | 1, 20 |
| 5 | Resonance | 21-27 | | | 21 |
| 6 | Aromaticity | 28-34 | 3, 18, 21-22 | 17-21 | |
| 7 | Stability of Resonating Structures | 35-37 | | 22-23 | 2 |
| 8 | Mesomeric Effect | 38-44 | 4-5, 19 | | 3, 17, 22 |
| 9 | Hyperconjugation | 45-63 | 6, 23-24 | 24-29 | 4 |
| 10 | Stability of Intermediates Part 1 | 64-69 | 25 | 30 | 5 |
| 11 | Stability of Intermediates Part 2 | 70-76 | 7-11 | 31-39 | 6-8 |
| 12 | Bond Length, Acid Base Equilibrium & Extraction of Compounds | 77-81 | 12 | 40-48 | 9-10, 23 |
| 13 | ACC | | 13-17, 20 | 49-50 | 11-16, 18-19 |

## Exercise - 1 Concept Builder

**SINGLE CORRECT TYPE**

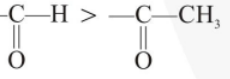
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Inductive effect

**Q1.** Which of the following is false for order of –I effect :-

(A) –F > –Cl > –Br > –I

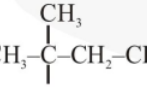
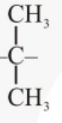
(B) —C≡CH > —CH=CH$_2$ > —CH$_2$—CH$_3$

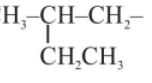
(C)  > 

(D)  > —C≡CH > H

**Q2.** Express in decreasing order of (+I) :

(a) $CH_3CH_2$–$CH_2$– (b) $CH_3$–

(c)  (d) 

(e) 

Correct answer is :-

(A) (c) > (d) > (e) > (a) > (b)

(B) (d) > (a) > (b) > (c) > (e)

(C) (a) > (b) > (c) > (d) > (e)

(D) (a) > (b) > (c) > (e) > (d)

**Q3.** Which of the following is correct order of –I power :

(A) $-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}H_3 > -\overset{\oplus}{N}Me_3 > -NO_2$

(B) $-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}Me_3 > -\overset{\oplus}{N}H_3 > -NO_2$

(C) –Cl > –F > –Br > –I

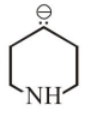
(D) $-\overset{\oplus}{N}H_3$ > –CN > $-NO_2$ > –F

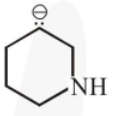
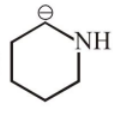
**Q4.** The inductive effects of the group $-CH_3$, $-CO\overset{\oplus}{O}^-$, –Br, $-NH_3^+$ respectively are

(A) +I, –I, +I, +I (B) +I, +I, –I, –I

(C) –I, –I, +I, +I (D) –I, +I, –I, +I

**Q5.** Which of the following is most stable :-

(A)  (B) 

(C)  (D) 

**Q6.** Which group does not exert [–I] effect :-

(A) $-\overset{+}{N}H_3$ (B) $-\overset{+}{N}R_3$

(C) $-NO_2$ (D) $-CR_3$

**Q7.** The correct order of increasing stability of the following carbocations is :-

(A) $CH_3\underset{F}{\underset{|}{C}}H\overset{+}{C}H_2 > FCH_2CH_2\overset{+}{C}H_2 > CH_3CH_2\overset{+}{C}H_2 > CH_3O\overset{+}{C}H_2$

(B) $CH_3O\overset{+}{C}H_2 > CH_3CH_2\overset{+}{C}H_2 > FCH_2CH_2\overset{+}{C}H_2 > CH_3\underset{F}{\underset{|}{C}}H\overset{+}{C}H_2$

(C) $FCH_2CH_2\overset{+}{C}H_2 > CH_3O\overset{+}{C}H_2 > CH_3CH_2\overset{+}{C}H_2 > CH_3\underset{F}{\underset{|}{C}}H\overset{+}{C}H_2$

(D) $CH_3\underset{F}{\underset{|}{C}}H\overset{+}{C}H_2 > CH_3CH_2\overset{+}{C}H_2 > FCH_2CH_2\overset{+}{C}H_2 > CH_3O\overset{+}{C}H_2 >$

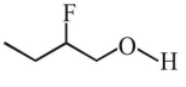
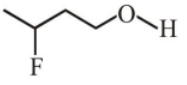
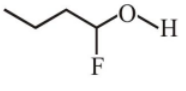
**Q8.** Pair of groups exerting (–I) effect is :-
(A) $-NO_2$ and $-CH_3$
(B) $-NO_2$ and –Cl
(C) –Cl and $-CH_3$
(D) $-CH_3$ and $-C_2H_5$

**Q9.** Which of the following statements is correct?
(A) +I group stabilised carbocation
(B) +I group destabilised carbocation
(C) –I group stabilised carbocation
(D) –I group destabilises stabilised carbanion

**Q10.** Arrange the following in their decreasing order of acidity :
(I) $CH_3OH$
(II) $CH_3CH_2OH$
(III) $(CH_3)_2CH–OH$
(IV) $(CH_3)_3C–OH$
(A) II > I > III > IV (B) I > II > III > IV
(C) IV > III > II > I (D) IV > III > I > II

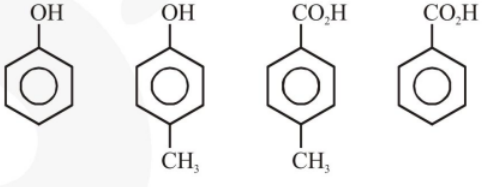
**Q11.** Decreasing(-I) power of given groups is :-
(A) –CN
(B) $-NO_2$
(C) $-NH_2$
(D) –F
(A) B > A > D > C (B) B > C > D > A
(C) C > B > D > A (D) C > B > A > D

## Acidity and basicity

**Q12.** In which of the following compounds is hydroxylic proton the most acidic ?
(A) 
(B) 
(C) 
(D) 

**Q13.** Increasing order of basic strength of following compounds is :-
(i) $CH_3–CH_2–NH_2$
(ii) $CH_3–CH=NH$
(iii) $CH_3–C\equiv N$
(A) (i) > (ii) > (iii)
(B) (ii) > (iii) > (i)
(C) (i) > (iii) > (ii)
(D) (iii) > (ii) > (i)

**Q14.** Correct acidic strength of followings :-
(I) $Cl_3C–CH_2–OH$
(II) $Cl_2CH–CH_2–OH$
(III) $Cl_2CH–CH_2–CH_2–OH$
(IV) $ClCH_2–CH_2–CH_2–OH$
(A) III > II > I > IV
(B) IV > III > II > I
(C) II > I > IV > III
(D) I > II > III > IV

**Q15.** 

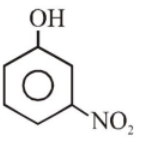
(a) (b) (c) (d)

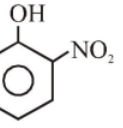
Compare acidic strength :-
(A) a > c > d > b
(B) d > a > c > b
(C) d > c > a > b
(D) d > c > b > a

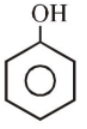
**Q16.** The correct order of acidic strength is :-
(A) Chloroacetic acid > Fluoroacetic acid > Phenol > Ethanol
(B) Ethanol > Phenol > Chloroacetic acid > Fluoroacetic acid
(C) Fluoroacetic acid > Chloroacetic acid > phenol > Ethanol
(D) Fluoroacetic acid > Chloroacetic acid > Ethanol > Phenol

**Q17.** Which is most acidic :-

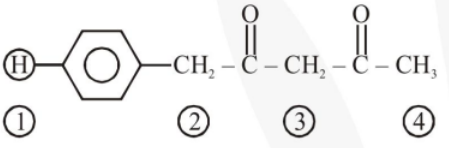
(A) 

(B) 

(C) 

(D) 

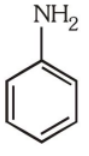
**Q18.** Which of the labeled hydrogen atom in the following structure is the most acidic

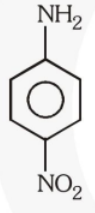


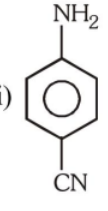
(A) 1 (B) 2

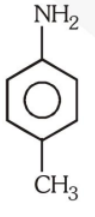
(C) 3 (D) 4

**Q19.** Arrange these compounds in decreasing order of basic strength :

(i) 

(ii) 

(iii) 

(iv) 

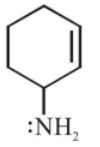
(A) (i) > (ii) > (iii) > (iv)

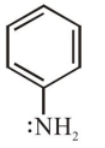
(B) (ii) > (iii) > (i) > (iv)

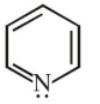
(C) (iv) > (i) > (iii) > (ii)

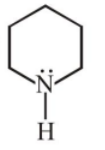
(D) (iv) > (i) > (ii) > (iii)

**Q20.** In which of following is minimum basic :-

(A) 

(B) 

(C) 

(D) 

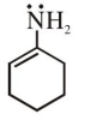
## Resonance

**Q21.** Which compound will not show the resonance?

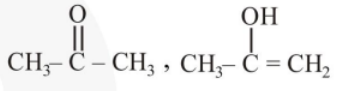
(A) 

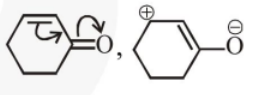
(B) 

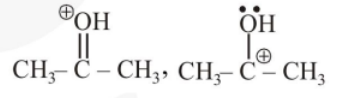
(C) 

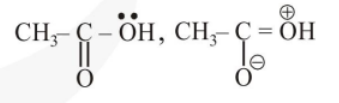
(D) 

**Q22.** Which of the following pairs of structures do not represent resonating structures -

(A) 

(B) 

(C) 

(D) 

**Q23.** Which of the following statement is incorrect?

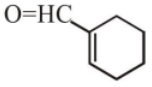
(A) Resonating structure are real & have real existence.

(B) Equivalent contributing structures make resonance hybrid very stable.

(C) Contributing structures are hypothetical having no real existance.

(D) Contributing structures are less stable than the resonance hybrid.

**Q24.** Which of the following compound has lone pair → π–conjugation :-

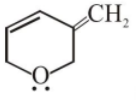
(A) 
(B) 

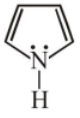
(C) 

(D) 

**Q25.** In which of the following cases lone-pair is involved in Resonance :-

I. 

II. 

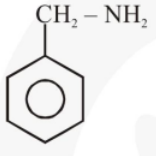
III. 

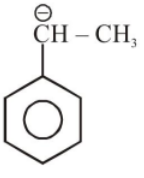
IV. $CH_2=CH-\ddot{O}-CH=CH_2$

(A) I, III (B) II, IV
(C) III, IV (D) I, II

**Q26.** Which of the following compounds has localized lone pair?

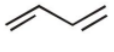
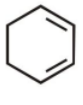
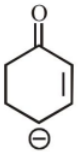
(A) 

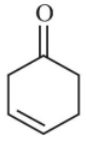
(B) 

(C) 

(D) 


**Q27.** In which of the following compound resonance does not takes place ?

(A) 
(B) 
(C) 

(D) 


## Aromaticity

**Q28.** Aromatic compound are these that meet the following criteria :

(i) The structure must be cyclic, containing $(4n + 2)\pi$ electrons
(ii) Each atom in the ring must have unhybridized p-orbitals
(iii) Unhybridized p-orbital must overlap
(iv) Structure must be planar

In anti-aromatic compound $(4n\pi)$ electron are delocalization, but delocalization of pi electrons over the ring increase the electronic energy.

Which of the following is anti aromatic?

(A) 
(B) 
(C) 

(D) All

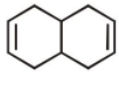
**Q29.** Which of the following is an aromatic species :

(A) 
(B) 
(C) 
(D) 


**Q30.** Among the following aromatic compound is

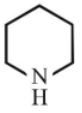
(A) 
(B) 
(C) 
(D) 

**Q31.** Which of the following will show aromatic behaviour :-

(A) 
(B) 
(C) 
(D) 

**Q32.** Which of the following is heterocyclic aromatic species :-

(A)  (B) 

(C)  (D) 

**Q33.** Which of following is non Aromatic :-

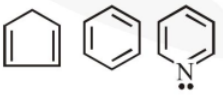
(A)  (B) 

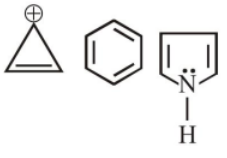
(C)  (D) 

**Q34.** Which of the following set have only aromatic compounds ?

(A) 

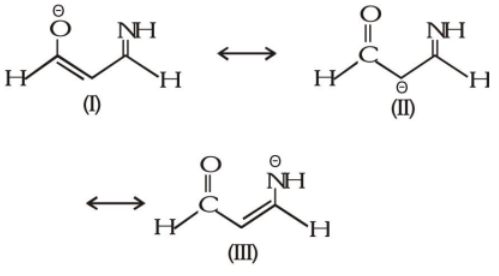
(B) 

(C) 

(D) 

## Stability of Resonating structure

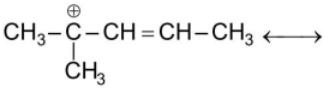
**Q35.** The correct order of stability among following canonical structures is :

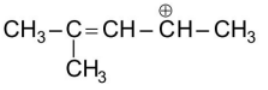


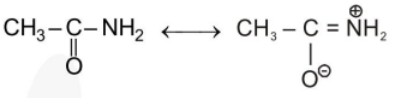
(A) I > II > III (B) I > III > II

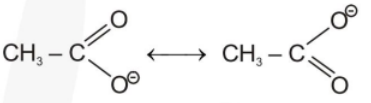
(C) II > I > III (D) III > I > II

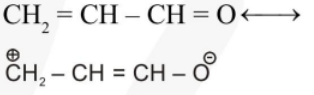
**Q36.** Which of the following resonating structure contributes equally to the resonance hybrid.

(A) 

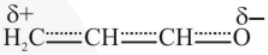


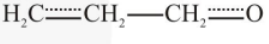
(B) 

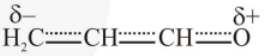
(C) 

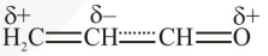
(D) 

**Q37.** Which is correct representation of resonance hybrid of $H_2C{=}CH{-}CH{=}O$.

(A) 

(B) 

(C) 

(D) 

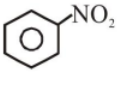
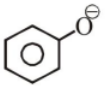
## Mesomeric Effect

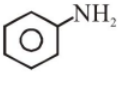
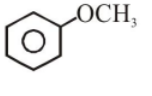
**Q38.** Which of the following is not **o**, **p**-directing:-

(A) – OH (B) – $CH_3$

(C) – Cl (D) – $NO_2$

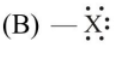
**Q39.** In which of the following molecules π-electron density in ring is maximum ?

(A)  (B) 

(C)  (D) 

**Q40.** Which one shows +M effect :-

(A) 

(B) 

(C) 

(D) 

**Q41.** Which of the following series contains atoms groups having only –M effect :-

(A) COR, OR, COOR

(B) Cl, CHO, $NH_2$

(C) $NO_2$, CN, $SO_3H$

(D) OH, $NR_2$, SR

**Q42.** Toluene is more reactive than benzene towards electrophilic substitution due to :-

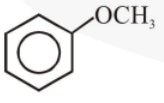
(A) Inductive effect only

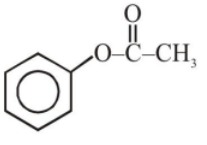
(B) Hyperconjugative effect only

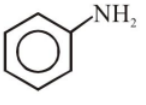
(C) Both inductive as well as hyperconjugative effects

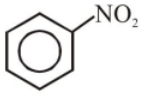
(D) Strong mesomeric effect

**Q43.** Which of the following group has (–M) effect :-

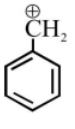
(A) 

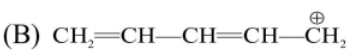
(B) 

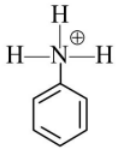
(C) 

(D) 

**Q44.** Which of the following compounds will not show resonance :-

(A) 

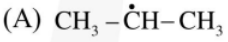
(B) 

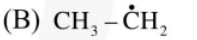
(C) 

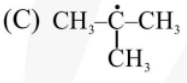
(D) 

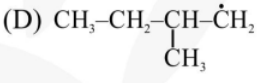
## Hyperconjugation

**Q45.** Which of the following free radical is most stable?

(A) 

(B) 

(C) 

(D) 

**Q46.** Correct order of heat of combustion is :

(p) 1, 3–pentadiene

(q) 1,3-Butadiene

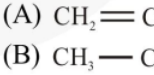
(r) 2,3–Dimethyl-1,3-butadiene

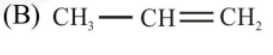
(s) Propadiene

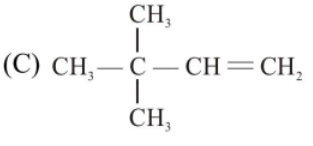
(A) s > q > p > r (B) s > r > q > p

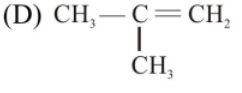
(C) r > p > q > s (D) s > r > p > q

**Q47.** Which of the following molecule has longest C = C bond length ?

(A) 

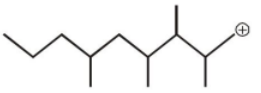
(B) 

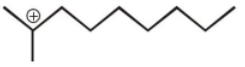
(C) 

(D) 

**Q48.** Select the most stable carbonacation among the following :-

(A) 

(B) 

(C) 

(D) 

**Q49.** Which of the following alkene is the most stable:-

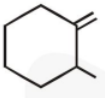
(A) 

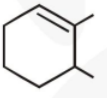
(B) 

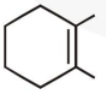
(C) 

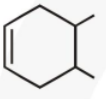
(D) 

**Q50.** Which of the following alkene has minimum heat of hydrogenation :

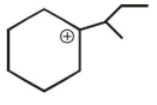
(A) 

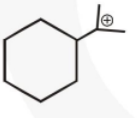
(B) 

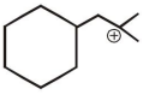
(C) 

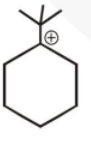
(D) 

**Q51.** Which cation is most stable :

(A) 

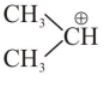
(B) 

(C) 

(D) 

**Q52.** Which of the following can not exhibit hyperconjugation :

(A) $CH_3-\overset{\square}{C}H_2$

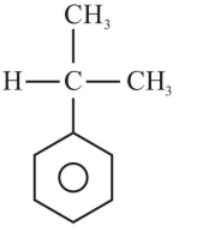
(B) 

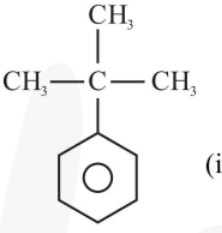
(C) $CH_3-CH=CH_2$

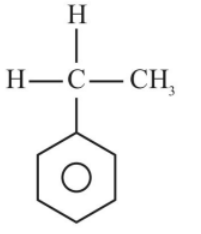
(D) $(CH_3)_3C-\overset{\oplus}{C}H_2$

**Q53.** Arrange following compounds in decreasing order of reactivity of ring towards attack of electron deficient species -

(i) 

(ii) 

(iii) 

(iv) 

(A) i > ii > iii > iv

(B) iii > iv > ii > i

(C) i > iv > ii > iii

(D) ii > ii > iv > iii

**Q54.** Most stable carbonium is :-

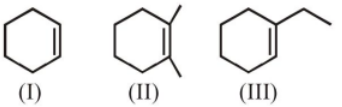
(A) $CH_3-CH_2-\overset{\oplus}{C}H_2$

(B) $CH_3-\overset{\oplus}{C}H-CH_2-CH_3$

(C) $CH_3-\overset{\oplus}{\underset{CH_3}{\underset{|}{C}}}-CH_3$

(D) $CH_3-\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{\overset{\oplus}{C}}}}}-CH_2$

**Q55.** Arrange the stability of following



(I) (II) (III)

(A) I < II < III

(B) II < I < III

(C) I < III < II

(D) II < III < I

**Q56.** Heat of combustion of which cycloalkanes is maximum

(A)  (B) 

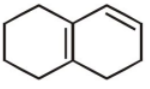
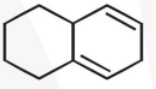
(C)  (D) 

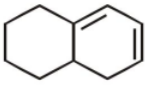
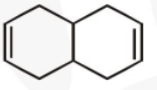
**Q57.** Which of following is maximum stable carbonium ion :-

(A) $\overset{\oplus}{C}H_3$ (B) $CH_3-\overset{\oplus}{C}H_2$

(C) $(CH_3)_2\overset{\oplus}{C}H$ (D) $(CH_3)_3\overset{\oplus}{C}$

**Q58.** Rank the following in decreasing order of heat of hydrogenation :-

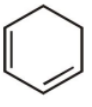
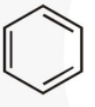
(i)  (ii) 

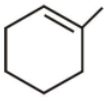
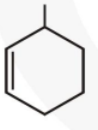
(iii)  (iv) 

(A) iii > i > ii > iv (B) i > iii > ii > iv

(C) ii > iv > iii > i (D) iv > iii > ii > i

**Q59.** Which is having maximum heat of hydrogenation?

(A)  (B) 

(C)  (D) 

**Q60.** Select the correct statement :

(i) Delocalisation of σ-electron is hyperconjugation.

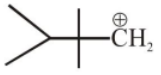
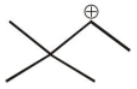
(ii) Delocalisation of π-electron is resonance.

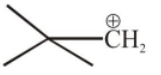
(iii) Permanent partial displacement of σ-electron is inductive effect.

(A) i & iii (B) ii & iii

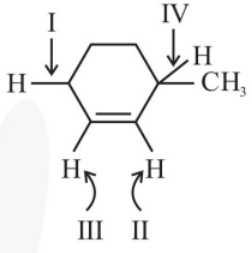
(C) i & ii (D) i, ii, iii

**Q61.** In which of the following compound hyperconjugation is observed ?

(A)  (B) 

(C)  (D) None of these

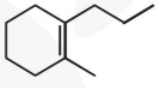
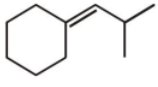
**Q62.** Which of the following C–H bonds participate in hyperconjugation ?

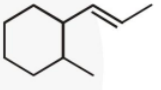
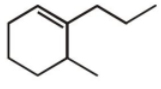



(A) I and II (B) I and IV

(C) I and III (D) III and IV

**Q63.** Which of the following shows minimum heat of combustion ?

(A)  (B) 

(C)  (D) 

## Stability of intermediates part 1

**Q64.** $\overset{\ominus}{C}H_3$ is less stable than :-

(A) $CH_3-\overset{\ominus}{C}H_2$

(B) $CH_3-\overset{\ominus}{C}H-CH_3$

(C) $\overset{\ominus}{C}H_2-NO_2$

(D) $CH_3-\overset{\ominus}{C}H-C_2H_5$

**Q65.** Which of the following statement is incorrect :-

(A) Inductive effect is distance dependent

(B) Inductive effect is permanent effect

(C) Resonance is an intramolecular process

(D) Phenol is more acidic than acetic acid

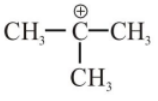
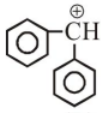
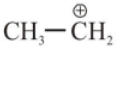
**Q66.** Which one of the following will give most stable carbocation on heterolytic cleavage of C–Cl bond :-

(A) $CH_2 = CH - Cl$

(B) $C_6H_5 - Cl$

(C) $CH_3 - CH = CH - Cl$

(D) $CH_2 = CH - CH_2 - Cl$

**Q67.**   

(A) (B) (C)

The relative stabilities of these carbocations are such that -

(A) (A) > (B) > (C)

(B) (B) > (C) > (A)

(C) (B) > (A) > (C)

(D) (C) > (B) > (A)

**Q68.** The correct order of stability of given anion is :-

(I) $CH_3–C\equiv C^{\ominus}$

(II) $CH_2(NO_2)–CH_2–O^{\ominus}$

(III) $(CH_3)_2\overset{\ominus}{C}–CD_3$

(IV) $(CH_3)_3C^{\ominus}$

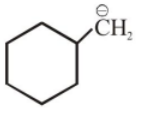
(A) II > I > III > IV

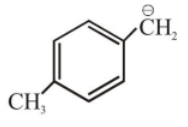
(B) I > II > IV > III

(C) II > I > IV > III

(D) I > II > III > IV

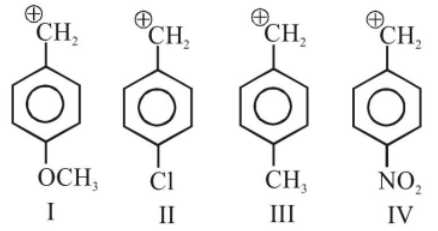
**Q69.** Which of the following is most stable anion?

(A)  (B) 

(C)  (D) 

## Stability of intermediates part 2

**Q70.** Which of the following is correct order of stability:-



I II III IV

(A) I > III > II > IV

(B) I > II > III > IV

(C) III > I > II > IV

(D) I > IV > III > II

**Q71.** Decreasing order of stability of

(a) $CH_3 - \overset{\oplus}{C}H - CH_3$

(b) $CH_3 - \overset{\oplus}{C}H - OCH_3$

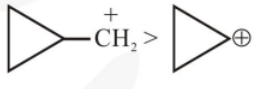
(c) $CH_3 - \overset{\oplus}{C}H - CO - CH_3$

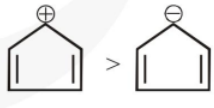
(A) a > b > c (B) b > c > a

(C) c > a > b (D) b > a > c

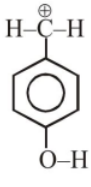
**Q72.** Correct stability order :-

(A) 

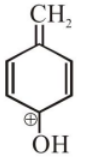
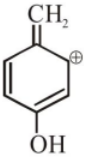
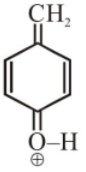
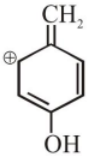
(B) 

(C) 

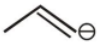
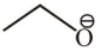
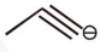
(D) All are correct

**Q73.**  (A).

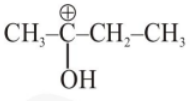
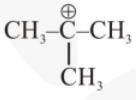
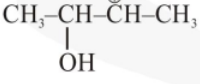
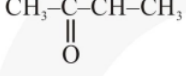
Most stable resonating structure for (A) is:-

(A) 
(B) 
(C) 
(D) 

**Q74.** Which is most stable anion ?

(A) 
(B) 
(C) 
(D) 

**Q75.** Which of the following carbocation is most stable?

(A) 
(B) 
(C) 
(D) 

**Q76.** Most stable carbanion is :

(A) 
(B) 
(C) 
(D) 

## Bond Legnth, acid base equillibrium & extraction of compounds

**Q77.** $CH_3O-CH=CH-NO_2$ I

$CH_2=CH-NO_2$ II

$CH_2=CH-Cl$ III

$CH_2=CH_2$ IV

Which of the following is the correct order of C – C bond lengths among these compounds :

(A) I > II > III > IV

(B) IV > III > II > I

(C) I > III > II > IV

(D) II > III > I > IV

**Q78.** In which of the following compounds carbon–chlorine bond distance is minimum:-

(A) $CH_3-Cl$

(B) $C_6H_5-CH_2-Cl$

(C) $CH_2=CH-Cl$

(D) $CH_2=CH-CH_2-Cl$

**Q79.** Which of the following shows the correct order of decreasing basicity in aqueous medium

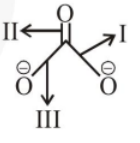
(A) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$

(B) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$

(C) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(D) $(CH_3)_2NH > CH_3NH_2 > NH_3 > (CH_3)_3N$

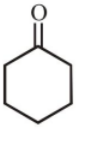
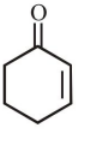
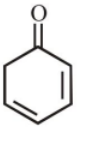
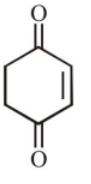
**Q80.** Correct order of bond length is :-



(A) I > II = III (B) II > III > I

(C) I > III > II (D) II = III = I

**Q81.** Which has maximum C–O bond length :-

(A) 
(B) 
(C) 
(D) 

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct* answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** Which of the following is true order of - I effect :-

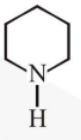
(A) $-F > Cl > -Br > -I$

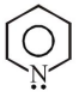
(B) $-\overset{+}{N}R_3 > -\overset{+}{N}H_3 > -NO_2$

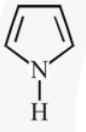
(C) $-F > -OH > -NH_3$

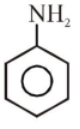
(D)  $> C \equiv CH > -H$

**Q2.** Which of the following less basic then



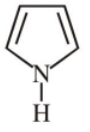
(A) 

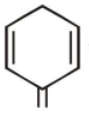
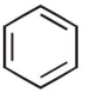
(B) 

(C) 

(D) None

**Q3.** In which of the following 2nd structure is aromatic ?

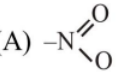
(A)  vs 

(B)  vs 

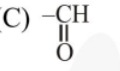
(C) $CH_2=CH-CH=CH-\overset{\ominus}{C}H_2$ vs 

(D)  V/S 

**Q4.** Groups which are showing -M effect :-

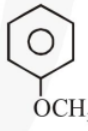
(A) 

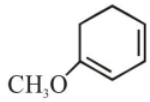
(B) 

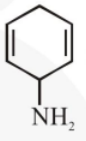
(C) 

(D) 

**Q5.** In which of the following + M effect is in operative mode -

(A) 

(B) 

(C) 

(D) All

**Q6.** Which of the following has Lesser HOH for per mole of $H_2$ than:-



(A) 

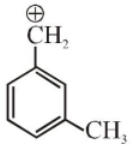
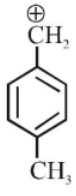
(B) 

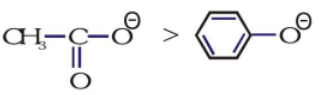
(C) 

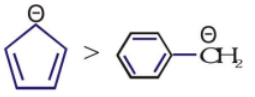
(D) None

**Q7.** Which are correct stability order :-

(A) $CH_3-CH=CH-CH_3 > CH_2=CH-CH_2-CH_3$

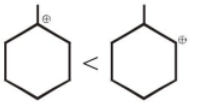
(B)  $>$ 

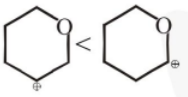
(C) 

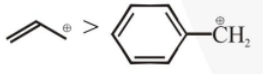
(D) 

**Q8.** Which are incorrect order of stability of carbocation :-

(A) 

(B) 

(C) 

(D) 

**Q9.** Correct options about I and II is/are



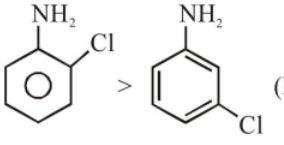
(A) I > II stability

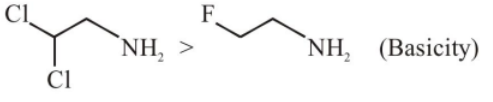
(B) At least one carbon is $sp^2$ in Both I & II.

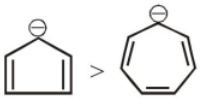
(C) At least one of carbon in case - I & II has Incomplete octet

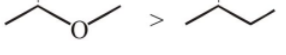
(D) All are correct

**Q10.** Which of the following is/are incorrect order:-

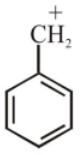
(A)  (Basic Strength)

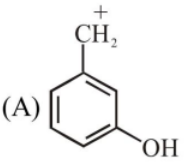
(B)  (Basicity)

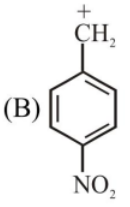
(C)  (Stability)

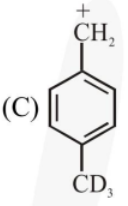
(D)  (Stability)

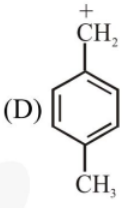
**Q11.** Which of the following more stable than -



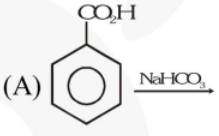
(A) 

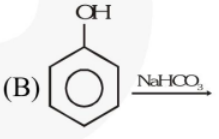
(B) 

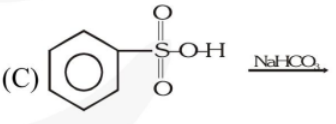
(C) 

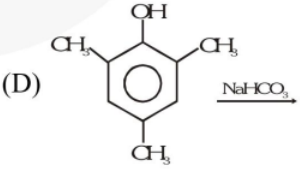
(D) 

**Q12.** In which of the following reaction $CO_2$ gas will evolve.

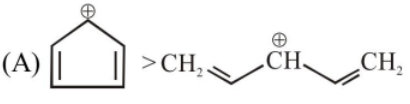
(A) 

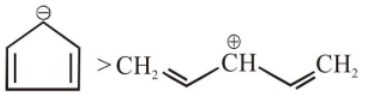
(B) 

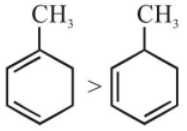
(C) 

(D) 

**Q13.** Compare the correct Resonance energy order :-

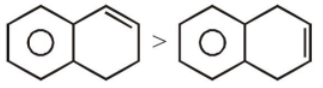
(A) 

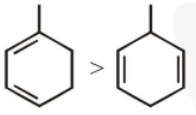
(B) 

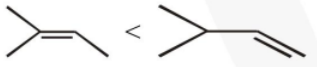
(C) 

(D) None

**Q14.** Which of the following correct order of stability :-

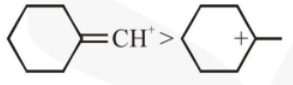
(A) 

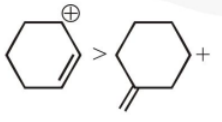
(B) 

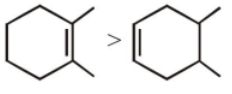
(C) 

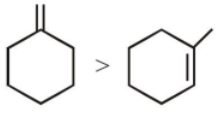
(D) All are correct

**Q15.** Which stability order is/are correct :-

(A) 

(B) 

(C) 

(D) 

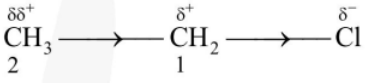
## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph for Q. 16 to 17**

When a covalent bond is formed between atoms of different electronegativity, the electron density is more towards the more electronegative atom of the bond. Such a shift of electron density results in a polar covalent bond. Bond polarity leads to various electronic effects in organic compounds.

Let us consider cholorethane ($CH_3CH_2Cl$) in which the C –Cl bond is a polar covalent bond. It is polarised in such a way that the carbon-1 gains some positive charge ($\delta^+$) and the chlorine some negative charge ($\delta^-$).



Such polarisation of σ-bond caused by the polarisation of adjacent σ-bond is referred to as the inductive effect.

**Q16.** Choose the incorrect option about inductive effect?

(A) Inductive effect of Cl decreases on terminal carbon of chloroethane as intervening bonds increases.

(B) Inductive effect is a distance dependent effect.

(C) $-CT_3$ is more electron donating by inductive than $-CH_3$.

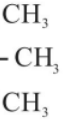
(D) Inductive effect is a temparory and weak effect.

**Q17.** In which of the following pair, both atom / group will show electron donating (+I) effect?

(A) $-CH_3$ , $-OCH_3$

(B) $-O^{\ominus}$ , $-CH_3$

(C) $-NO_2$ , $-F$

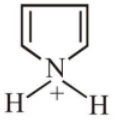
(D) $-OH$ , 

## MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

**Q18.** **List - I** **List - II**

(P)  (1) Aromatic

(Q)  (2) Non aromatic

(R)  (3) Unstable at room temperature because of antiaromatic character

(S)  (4) Homocyclic

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1 | 2 | 1,4 | 3,4 |
| (B) | 3,4 | 1,4 | 1 | 2 |
| (C) | 1 | 2 | 3 | 4 |
| (D) | 1 | 2 | 3,4 | 1,4 |

**Q19.** **Match the list:**

| **List-I (Group )** | **List-II (Effect)** |
|---|---|
| (P) $-CH_3$ | (1) + I |
| (Q) $-\overset{+}{N}H_3$ | (2) – I |
| (R) –CN | (3) + M / +R |
| (S) $-NH_2$ | (4) – M / –R |

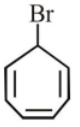
**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 2 | 2,4 | 1 | 2,3 |
| (B) | 1 | 2 | 2,4 | 2,3 |
| (C) | 1 | 2,4 | 2,3 | 2 |
| (D) | 2,4 | 2,3 | 1 | 2 |

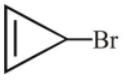
## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

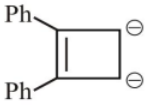
**Q20.** **Column I** **Column II**

(A) 

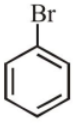
(P) Anti Aromatic

(B) 

(Q) Aromatic

(C) 

(R) Non-aromatic

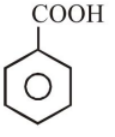
(D) 

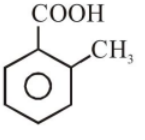
(S) Gives yellow ppt with $AgNO_3$

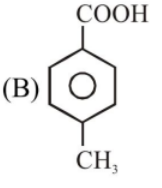
## NUMERICAL TYPE

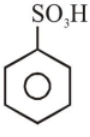
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

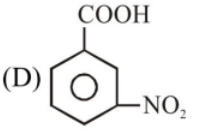
**Q21.** How many of the following are more Acidic than -

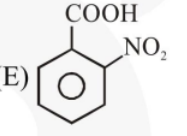


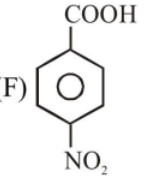
(A) 

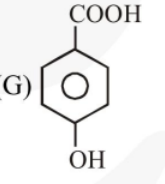
(B) 

(C) 

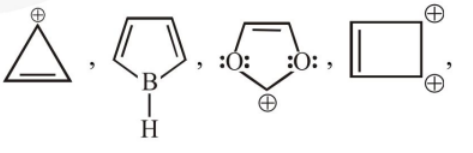
(D) 

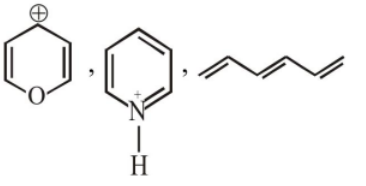
(E) 

(F) 

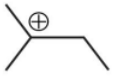
(G) 

**Q22.** How many of following Aromatic :-



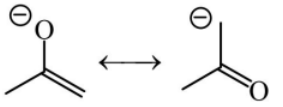


**Q23.** Identify total number of hyper conjugating structures involving C–H bond in following carbocation?



**Q24.** Total number of pairs of resonating structure in which second is more stable than first.

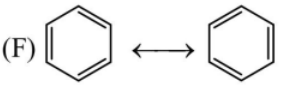
(A) $CH_3-\overset{\oplus}{C}=O \longleftrightarrow CH_3-C\equiv\overset{\oplus}{O}$

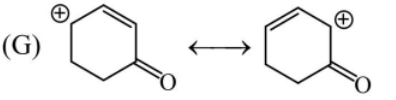
(B) 

(C) 

(D) 

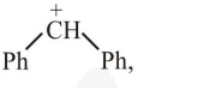
(E) 

(F) 

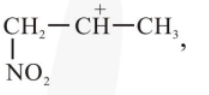
(G) 

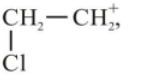
**Q25.** Number of carbocation Which is stable than

$CH_2(Cl)-\overset{+}{C}H_3-CH_3$ are -











## *Exercise - 2* — *Brain Booster*

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct* answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** Which of the following statement is incorrect for inductive effect ?

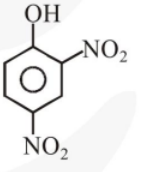
(A) It involves shifting of σ bonded electron

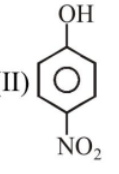
(B) It is distance dependent effect

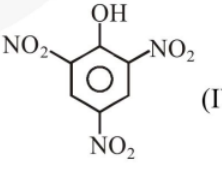
(C) It arise due to difference in electro-negativity of atom attach to organic chain

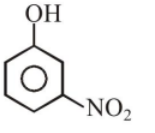
(D) It is a temporary effect

**Q2.** The correct order of increasing acid dissociation constant

(I) 

(II) 

(III) 

(IV) 

(A) II < IV < I < III

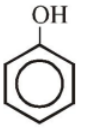
(B) IV< III < I < II

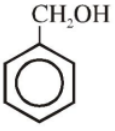
(C) IV < II < I < III

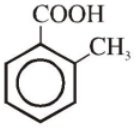
(D) IV < I < II < III

**Q3.** Which of the following is most acidic ?

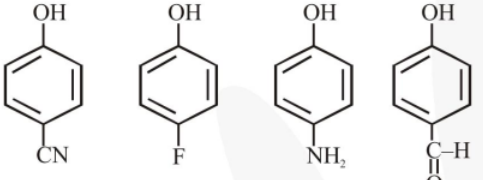
(A) 


(B) 


(C) 

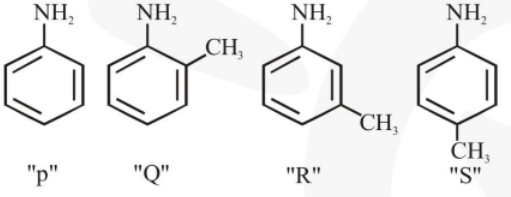

(D) 


**Q4.** The decreasing order of acidity of following phenol derivative is :-



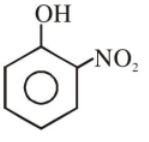

(A) P > S > Q > R (B) S > Q > P > R
(C) R > Q > P > S (D) P > Q > R > S

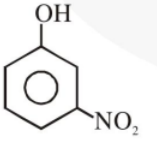
**Q5.** The decreasing order of basic strength of the following aniline derivative :-

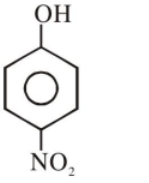


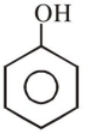

(A) S > R > Q > P (B) P > Q > R > S
(C) S > Q > P > R (D) S > R > P > Q

**Q6.** Order of Acidic strength :

(a) 

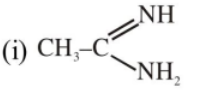

(b) 


(c) 


(d) 

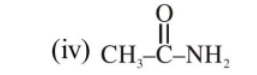

(A) a > b > c > d (B) c > a > b > d
(C) a > c > b > d (D) c > b > a > d

**Q7.** The correct order of basic strength :

(i) 

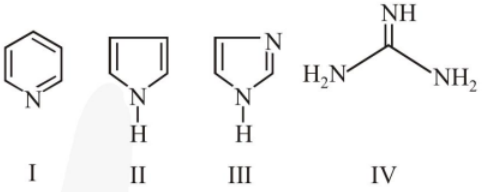

(ii) $CH_3–CH_2–NH_2$

(iii) $(CH_3)_2NH$

(iv) 

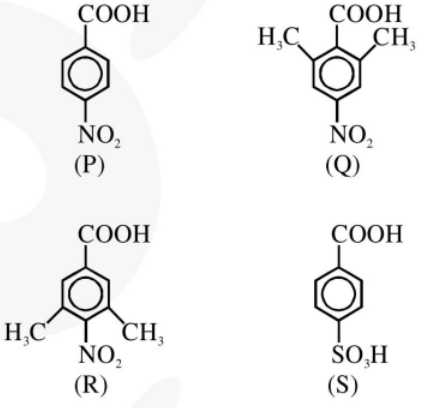

(A) ii > i > iii > iv (B) i > iii > ii > iv
(C) iii > i > ii > iv (D) i > ii > iii > iv

**Q8.** The correct order of basic strength.



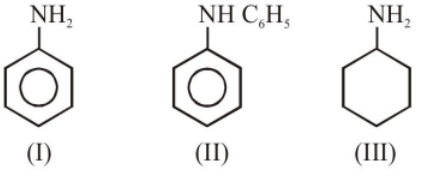

(A) IV > I > II > III (B) IV > I > III > II
(C) III > IV > I > II (D) IV > III > I > II

**Q9.** Decreasing order of acidic strength of following compound :-



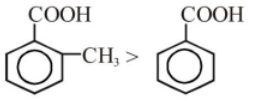

(A) P > Q > R > S (B) S > R > P > Q
(C) S > Q > P > R (D) Q > R > P > S

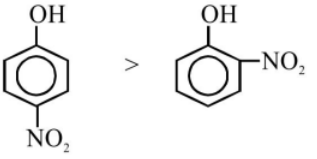
**Q10.** Arrange the following in decreasing kb value :-



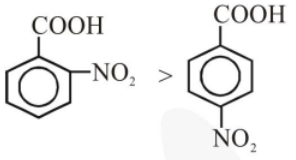

(A) III > I > II (B) III < I < II
(C) III < II < I (D) II < III < I

**Q11.** Which of following is correct for acidic strength:-

(A)  >

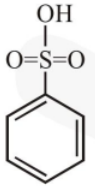
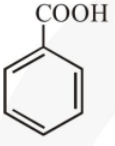
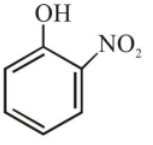
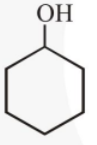
(B) 

(C) $CHCl_3 < CHF_3$

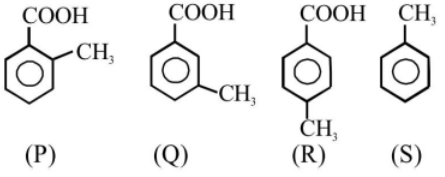
(D) 

(A) B and C
(B) C and D
(C) A, B and D
(D) A, B, and C

**Q12.** For which of the following compound Ka value will be maximum :-

(A) 
(B) 
(C) 
(D) 

**Q13.** The decreasing order of acidity of following benzoic derivatives is :-



(P) (Q) (R) (S)

(A) P > Q > R > S
(B) S > R > Q > P
(C) P > S > Q > R
(D) Q > S > P > R

**Q14.** Correct in decreasing order of $PK_b$ values:-

(A) $C_6H_5–NH–CH_3 > C_6H_5–NH_2 > (C_2H_5)_2NH > C_2H_5NH_2$

(B) $C_6H_5–NH–CH_3 > (C_2H_5)_2NH > C_2H_5–NH_2 > C_6H_5NH_2$

(C) $(C_2H_5)_2NH > C_2H_5–NH_2 > C_6H_5–NH_2 > C_6H_5–NH–CH_3$

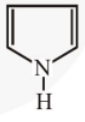
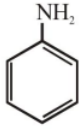
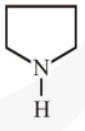
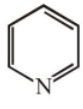
(D) $C_6H_5–NH_2 > C_6H_5–NH–CH_3 > C_2H_5–NH_2 > (C_2H_5)_2NH$

**Q15.** Identify the strongest base from the given compounds :-

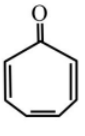
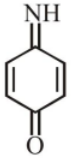
(A) 
(B) 
(C) 

(D) All have same basic strength

**Q16.** Strongest base among following is :-

(A) 
(B) 
(C) 
(D) 

**Q17.** Which is aromatic compound -

(A) 
(B) 
(C) 
(D) 

**Q18.** Which is not stable at room temperature?

(A)  (B) 

(C)  (D) 

**Q19.** Among the following is aromatic compounds :-

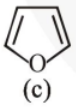
(a)  (b) 

(c)  (d) 

(A) a, b, c (B) a, b, c, d

(C) a, c, d (D) b, c, d

**Q20.** Which of the following is aromatic :-

 (a)  (b)  (c)  (d)

(A) a, b (B) b, c

(C) c, d (D) a, d

**Q21.** Which of the following compound is unstable at room temperature :-

(A)  (B) 

(C)  (D) 

**Q22.** Which will be the least stable resonating structure?

(A) $CH_2=CH-\overset{\oplus}{C}H-\overset{\ominus}{C}H-\ddot{N}H_2$

(B) $\overset{\ominus}{C}H_2-\overset{\oplus}{C}H-CH=CH-\ddot{N}H_2$

(C) $\overset{\ominus}{C}H_2-CH=CH-CH=\overset{\oplus}{N}H_2$

(D) $CH_2=CH-\overset{\ominus}{C}H-CH=\overset{\oplus}{N}H_2$

**Q23.** The most stable resonating structure of $CH_3-\ddot{O}-CH=CH_2$ is :-

(A) $CH_3-O-\overset{\ominus}{C}H-\overset{\oplus}{C}H_2$

(B) $CH_3-\overset{\oplus}{O}=CH-\overset{\ominus}{C}H_2$

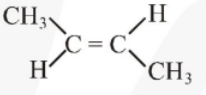
(C) $CH_3-O-\overset{\oplus}{C}H-\overset{\ominus}{C}H_2$

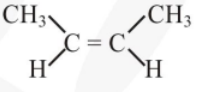
(D) All are equally stable

**Q24.** The relative stability of the compounds

(I) $CH_3-\overset{CH_3}{\overset{|}{C}}=\overset{CH_3}{\overset{|}{C}}-CH_3$

(II) $CH_3-\overset{CH_3}{\overset{|}{C}}=CH-CH_3$

(III) 

(IV) 

(V) $CH_3—CH=CH_2$

(VI) $CH_2=CH_2$

is in the order -

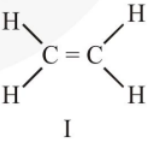
(A) I > II > III > IV > V > VI

(B) VI > V > IV > III > II > I

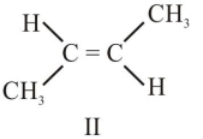
(C) I > III > V > II > IV > VI

(D) II > I > IV > III > V > VI

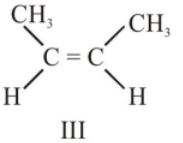
**Q25.** Correct stability order of following alkene will be:-



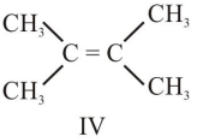
I



II



III



IV

(A) II > IV > III > I (B) IV > II > III > I

(C) III > IV > I > II (D) I > II > III > IV

**Q26.** Stability of following alkenes in the increasing order is :-

(I) $CH_3–CH=CH–CH_2–CH_3$

(II) $CH_3-\underset{\underset{CH_3}{|}}{C}=\underset{\underset{CH_3}{|}}{C}-CH_3$

(III) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2$

(IV) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$

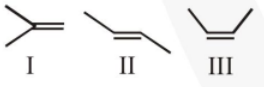
(A) I < III < IV < II

(B) I < II < III < IV

(C) IV < III < II < I

(D) III > I > IV > II

**Q27.** Correct statements about the following is :-




(A) I > II > III stability order

(B) III> II > I Heat of hydrogenation

(C) Both (A) & (B)

(D) None

**Q28.** Which of the following alkene has highest value of heat of hydrogenation ?

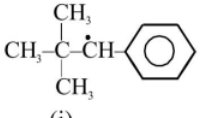
(A) 

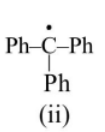
(B) 

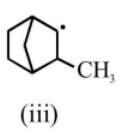
(C) 

(D) 

**Q29.** Hyperconjugation occurs in :-


(i)


(ii)


(iii)

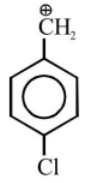
(A) Only (i)

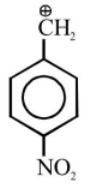
(B) (i) & (ii)

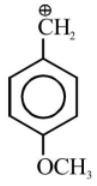
(C) (i), (ii) & (iii)

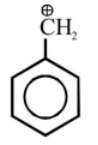
(D) only (iii)

**Q30.** Most stable carbocation is :-

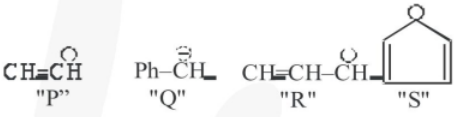
(A) 

(B) 

(C) 

(D) 

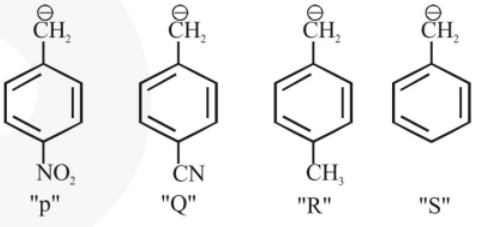
**Q31.** Arrange the following carbanions in decreasing order of stability:-




(A) P > Q > R > S (B) S > Q > P > R

(C) S > Q > R > P (D) Q > S > R > P

**Q32.** The decreasing order of stability of the following anion is:-




(A) S > R > Q > P (B) P > Q > R > S

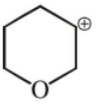
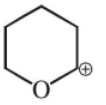
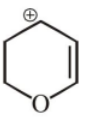
(C) Q > P > R > S (D) P > Q > S > R

**Q33.** Which is the least stable carbonium ion :-

(A) $(C_6H_5)_3\overset{\oplus}{C}$ (B) $C_6H_5\overset{\oplus}{C}H_2$

(C) $(CH_3)_2\overset{\oplus}{C}H$ (D) $CH_2=CH-\overset{\oplus}{C}H_2$

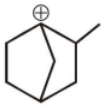
**Q34.** Order of stability of following carbocation:

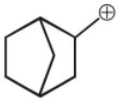
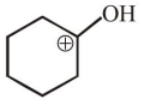
(a)  (b)  (c) 

(A) c > b > a (B) a > b > c

(C) b > c > a (D) c > a > b

**Q35.** Find out correct stability order in the following carbocations-

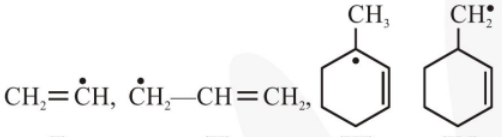
(I)  (II) 

(III)  (IV) 

(A) IV > I > III > II (B) IV > III > I > II

(C) I > IV > III > II (D) I > III > IV > II

**Q36.** Stability of following radical is :-



(I) (II) (III) (IV)

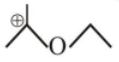
(A) II > III > II > IV

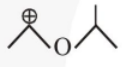
(B) III > II > I > IV

(C) III < II < I < IV

(D) I < IV < II < III

**Q37.** The correct stability order for the following carbocation is :-

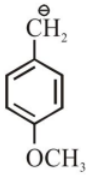
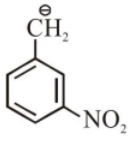
(I)  (II) 

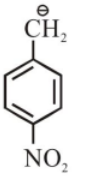
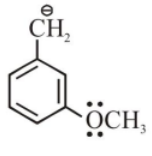
(III)  (IV) 

(A) IV > I > III > II (B) IV > III > II > I

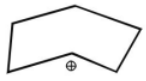
(C) I > IV > III > II (D) I > III > IV > II

**Q38.** Which of the following is most stable anion :-

(A)  (B) 

(C)  (D) 

**Q39.** Correct order of stability for the following -

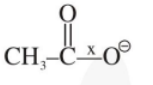
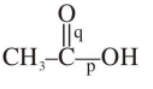
(I)  (II) 

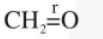
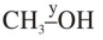
(III)  (IV) 

(A) IV > III > I > II (B) II > IV > III > I

(C) III > IV > I > II (D) None

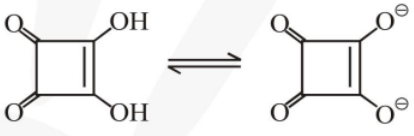
**Q40.** Correct order of bond length :

(A) y > x > p > q > r (B) y > p > x > q > r

(C) p > x > y > q > r (D) p > y > x > q > r

**Q41.** 

Squaric acid Squaric acid dianion

Squaric acid dianion has :

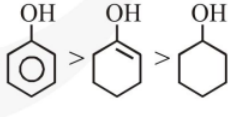
(A) 4-identical resonating structures

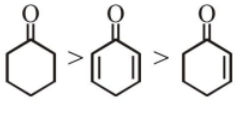
(B) C–C bond length is equal to C–O bond length

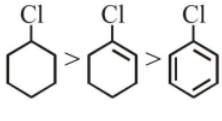
(C) 3 nonidentical resonating structures

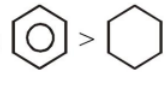
(D) Localised negative charged on oxygen

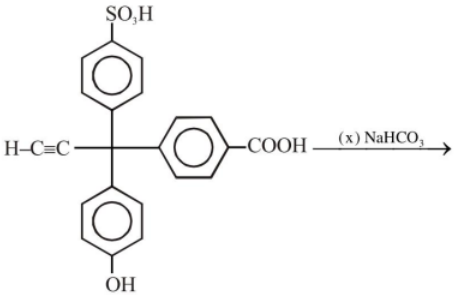
**Q42.** Which one is correct for bond length :-

(A)  [C–O] bond length

(B)  [C–O] bond length

(C)  [C–Cl] bond length

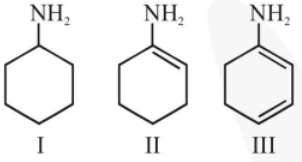
(D)  [C–C] bond length

**Q43.** 


Total number of moles of $NaHCO_3$ consumed during given reaction will be x, so the value of x will be :-

(A) 1 (B) 3 (C) 2 (D) 4

**Q44.** Compare C–N bond length in the following:



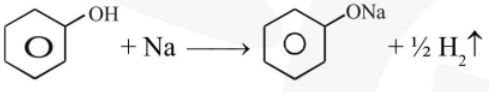

Correct order is -

(A) I > II > III (B) I > III > II

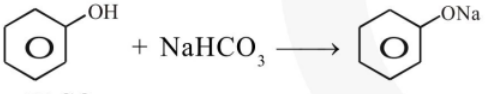
(C) III > II > I (D) III > I > II

**Q45.** Which of the following is incorrect:

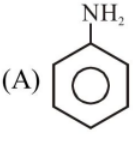
(A) $R–COOH + Na \rightarrow R–COONa + ½\ H_2\uparrow$

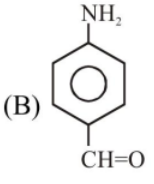
(B) 

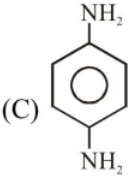

(C) $R–COOH + NaHCO_3 \longrightarrow R–COONa + H_2O + CO_2\uparrow$

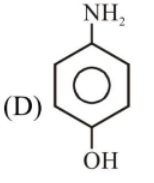
(D) 


**Q46.** C–N bond length is minimum in :-

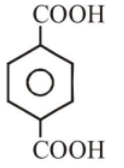
(A) 

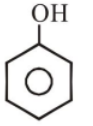

(B) 

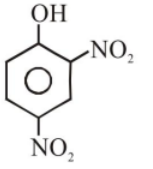

(C) 

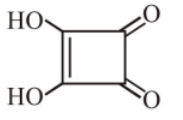

(D) 


**Q47.** Which of the following not give $CO_2$ gas with $NaHCO_3$ ?

(A) 

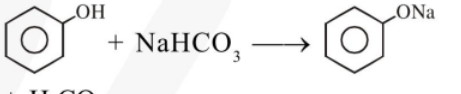

(B) 


(C) 


(D) 


**Q48.** Which of the following acid base reaction is least feasible :

(A) $CH \equiv CH + NaNH_2 \longrightarrow CH \equiv CNa + NH_3$

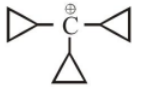
(B) 


(C) $CH_3–COOH + NaOH \longrightarrow CH_3–COONa + H_2O$

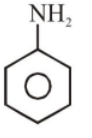
(D) $CH_3–CH_2–OH + CH \equiv CNa \longrightarrow CH_3–CH_2–ONa + CH \equiv CH$

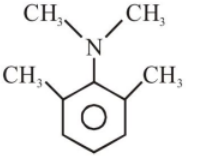
**Q49.** Which one amongst the following carbocations is most stable ?

(A) $C_6H_5–\overset{\oplus}{C}H–C_6H_5$ (B) 

(C) $CH_3–\overset{\oplus}{C}(CH_3)–CH_3$ (D) 

**Q50.** In which of the following lone pair involved in Resonance :-

(A) 

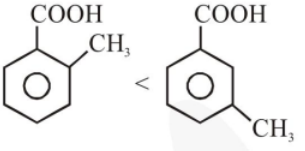

(B) 

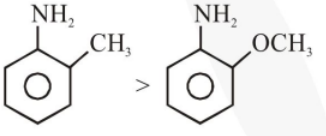

(C) Both (A) & (B) (D) None of these

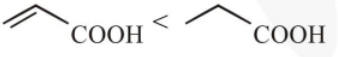
## MORE THAN ONE CORRECT TYPE

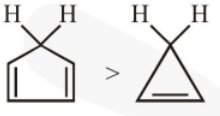
*Each question has FOUR options for correct answer(s).* ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** Incorrect order of Acidity :-

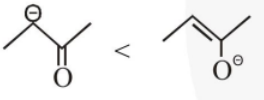
(A) 

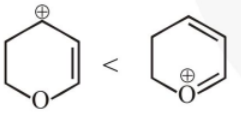
(B) 

(C) 

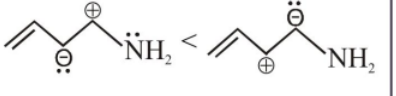
(D) 

**Q2.** Which of the following are correct :-

(A) Stability 

(B) Stability 

(C) Stability 

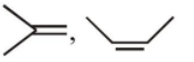
(D) Stability 

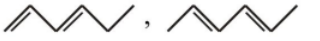
**Q3.** Which of the following can show + M effect-

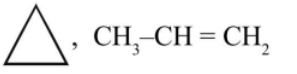
(A) $-OCH_3$ (B) $-NHCOCH_3$

(C) $-\overset{\overset{O}{\|}}{C}-OCH_3$ (D) $-O^-$

**Q4.** Select the correct pair in which 1st has more HOH than 2nd

(A) 

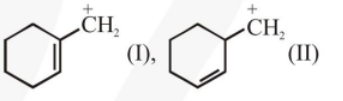
(B) 

(C) 

(D) None

**Q5.** In which of the following cases, the carbocation (I) are more stable than the carbocation (II) ?

(A) $C_6H_5-\overset{+}{C}H_2(I), CH_2=CH-\overset{+}{C}H_2(II)$

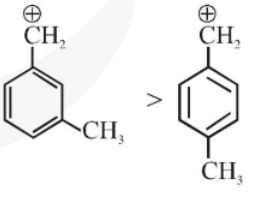
(B) 

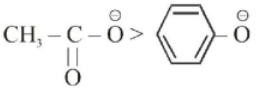
(C) $CH_2=\overset{+}{C}H(I), CH_3-\overset{+}{C}H_2(II)$

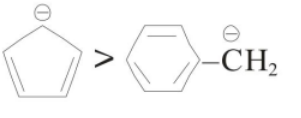
(D) $H_3C-\overset{+}{C}H_2(I), \underset{F}{CH_2}-\overset{\oplus}{C}H_2(II)$

**Q6.** Which are correct stability order :-

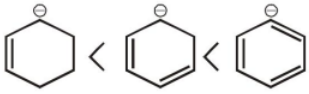
(A) $CH_3-CH=CH-CH_3 > CH_2=CH-CH_2-CH_3$

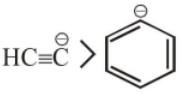
(B) 

(C) 

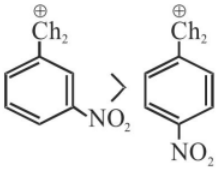
(D) 

**Q7.** Which are correct order of stability :-

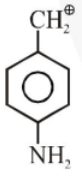
(A) 

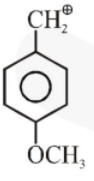
(B) 

(C) $CH_3-\overset{\oplus}{C}H-CH_3 > CH_3-\overset{\oplus}{C}H-CH_2-CH_3$

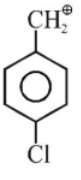
(D) 

**Q8.** Which carboanion is/are less stable than :



(A) 

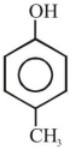
(B) 

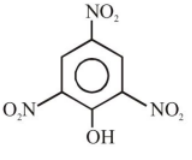
(C) 

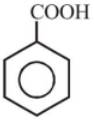
(D) None

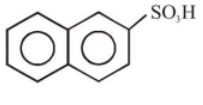
**Q9.** How many of the following will produce effervescence of $CO_2$ with $NaHCO_3$

(a) Me–COOH (b) $Ph-SO_3H$

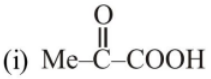
(c) 

(d) 

(e) 

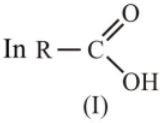
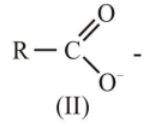
(f) 

(g) $CH_3-CH_2-OH$ (h) $CH_3-C\equiv CH$

(i) 

(A) 6 (B) 7

(C) 8 (D) 9

**Q10.** In  and  -

Incorrect statement about I & II -

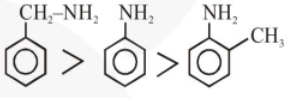
(A) In case of I C = O and C - O bond length identical

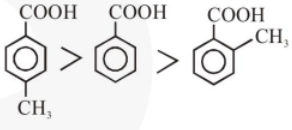
(B) In case - II C = O and C - O bond length identical

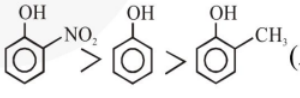
(C) Both in (I) & (II) C = O bond length same

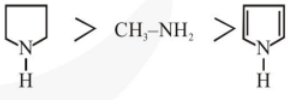
(D) (B) and (C) are correct

**Q11.** What are the correct order :

(A)  (Basic strength)

(B)  (Acidic strength)

(C)  (Acidic strength)

(D)  (Basic strength)

**Q12.** (A)  (B)  (C) 

Choose the correct statement -

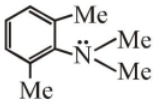
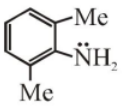
(A) A is more basic than B

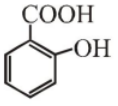
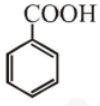
(B) B is more basic than A

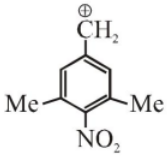
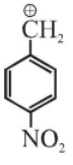
(C) B is more basic than C

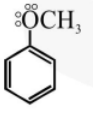
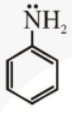
(D) All are aromatic bases

**Q13.** Which are correct statement :-

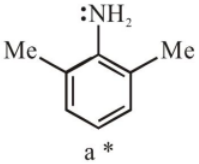
(A)  is more basic than 

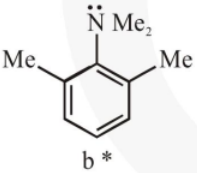
(B)  is more acidic than 

(C)  is more stable than 

(D)  is more reactive than 

towards E.S.R.

**Q14.** 
a *
I


b *
II

Correct statement about I & II -

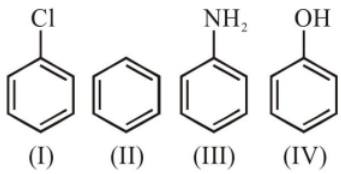
(A) Electron density at Ia* > IIb*

(B) No. of Hydrogen that can show Hyper conjugation is same in both.

(C) Rate of effective resonance between –N̈ lone pair and ring is same.

(D) In Both I and II $-NH_2$ and $-NMe_2$ can acts - I effects group.

**Q15.** Correct order of electron density in Benzene ring of following :-


(I) (II) (III) (IV)

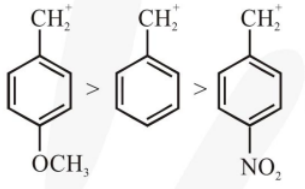
(A) I > II > III > IV

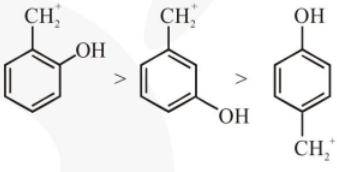
(B) II > I > III > IV

(C) IV > III > II > I

(D) III > IV > II > I

**Q16.** Which of th following is/are correct example :-

(A)  (Electron density at Benzene ring.)

(B)  (Stability Order)

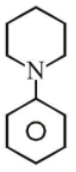
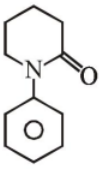
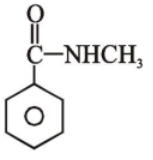
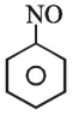
(C)  (No. of α - H)

(D)  (Stability Order)

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q17.** Match the Column

| **Column I** | **Column II** |
|---|---|
| (A)  | (P) –M group present |
| (B)  | (Q) +I group present |
| (C)  | (R) –I group present |
| (D)  | (S) +M group present |

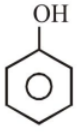
**Q18.** Match the Column

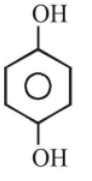
| **Column I** | **Column II** |
|---|---|
| (A) $C_6H_5CH_2^{\ominus}$ | (P) Pyramidal |
| (B)  | (Q) 2p-3d-Back Bonding |
| (C)  | (R) $sp^2$ hybridised |
| (D) $:CCl_2$ | (S) Aromatic |
| | (T) None |

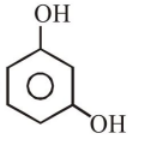
## NUMERICAL TYPE

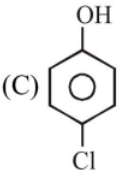
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

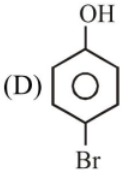
**Q19.** How many are more acidic than :-

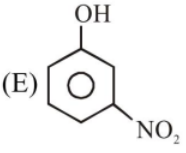


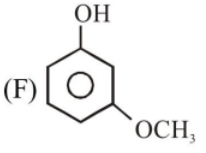
(A) 

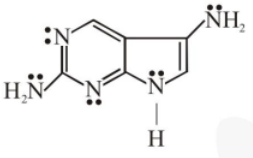
(B) 

(C) 

(D) 

(E) 

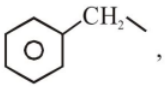
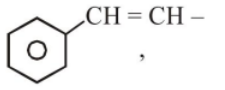
(F) 

**Q20.** 

Number of localised lone pair in heterocyclic compound is/are.

**Q21.** How many of the folowing can't participate in Resonance.

$-OH, -CH_2-CH_2^-, -CH_2-CH_2$

 ,  ,

$-BCl_2$, $-\overset{+}{P}Ph_3$, $-CF_3$

## Answer Key

| Ex-1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| D | A | B | B | D | D | B | B | A | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | D | A | D | C | C | A | C | C | B |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| B | A | A | C | C | B | D | A | A | A |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| B | C | D | D | B | C | A | D | B | B |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| C | C | D | C | C | C | D | C | D | C |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| C | D | C | C | C | D | D | C | B | D |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| B | B | A | C | D | D | C | C | C | A |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| D | B | C | C | A | A | A | C | C | D |
| 81 | | | | | | | | | |
| C | | | | | | | | | |

## Ex-1A

**1.** A, B, C **2.** A, B, C **3.** A, B, C **4.** A, C, D **5.** A, B

**6.** B, C **7.** A, C, D **8.** A, B, D **9.** B, C **10.** A, B

**11.** C, D **12.** A, C **13.** B, C **14.** A, B **15.** B, C

**16.** D **17.** B **18.** D **19.** B

**20.** A→R,S ; B→R,S ; C→Q ; D→Q **21.** 5

**22.** 5 **23.** 008 **24.** 4 **25.** 3

**Ex-2**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| D | C | D | A | D | B | B | D | C | A |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| C | A | C | D | A | C | C | C | C | B |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | A | B | A | B | A | C | D | D | C |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| C | D | C | A | A | D | C | C | A | B |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| A | C | C | A | D | B | B | B | D | A |

## Ex-2A

**1.** A, C **2.** A, B, C **3.** A, B, D **4.** B, C **5.** A, B, D

**6.** A, C, D **7.** B, C, D **8.** A, B, C **9.** A **10.** A, C, D

**11.** A, C, D **12.** B, C, D **13.** A, B, C **14.** A, B, D **15.** D

**16.** A, C, D

**17.** A→R,S ; B→R,S ; C→P,R ; D→P,R,S

**18.** A→R,S ; B→T ; C→R,S ; D→Q,R

**19.** 5 **20.** 2 **21.** 4